

MUMBA AMOS NTAMBO

AI Engineer | Full-Stack and Cloud



PERSONAL DATA

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Date of Birth: 29.04.2002 | **Nationality:** Zambian

EXPERIENCE

AI / Full-Stack Engineer (Freelance) | Baseline Health AI Platform, Kuala Lumpur, Malaysia

Healthcare AI start-up: Patient analytics and reporting platform

June 2025 to Current

[baseline.website](#)

- Built a women's health platform end to end: Next.js patient and clinic portals with a FastAPI service and Supabase Postgres, deployed on Vercel.
- Used GPT-4o vision for OCR and structured extraction of biomarker data from scanned lab reports, replacing manual entry of hormone panels and cutting report turnaround time by around 80%.
- Built a conversational patient intake chatbot on GPT-4o-mini, replacing static form entry and feeding structured data straight into the reporting pipeline.
- Generated clinical narratives with GPT-5 using structured outputs against a strict JSON schema, with all statistics computed in code so the model narrates figures rather than producing them.
- Built a two-store RAG layer on pgvector (HNSW cosine, text-embedding-3-small) giving each generated report patient-scoped retrieval over prior analyses and medical documents.
- Measured cross-source similarity as non-discriminating (0.043 vs 0.157 score spread) and switched that path to recency selection, cutting an embedding call per generation.
- Made retrieval fail-soft so embedding or RPC errors degrade output instead of blocking generation.
- Split model selection by cost and capability rather than defaulting to a single model: GPT-4o for vision extraction, GPT-5 for narratives, GPT-4o-mini for chat and first-pass triage, and text-embedding-3-small for retrieval.
- Built the access-control layer for a multi-tenant clinical platform: bearer-token identity resolved server-side on every request, with patient-facing routes deriving the patient from the token rather than any client-supplied ID.
- Scoped clinic staff to their own clinic's patients through a shared authorization module with read-only enforcement, and served medical files exclusively through one-hour signed URLs from private buckets, never public URLs.

Tech: Python, FastAPI, Next.js, OpenAI GPT-4o/GPT-5, pgvector, Supabase, Vercel, PostgreSQL

DevOps / Platform Engineer (Freelance) | GoAcara Event Management Platform, Kuching, Malaysia

Microservices-based event management SaaS

August 2025 to January 2026

[goacara.com](#)

- Provisioned a 4-node EKS cluster on AWS with cluster auto-scaling and ALB ingress, running a microservices event platform in production.
- Wrote the full infrastructure in Terraform with remote state in S3, so the environment could be torn down and rebuilt from code rather than reconstructed by hand.
- Built GitOps delivery with ArgoCD and GitHub Actions, so deployments came from committed manifests rather than manual kubectl.
- Containerised services with Podman and Docker to keep local development and production builds identical, removing environment drift as a class of bug.
- Ran the platform end to end as sole engineer: cluster, pipelines, database, and application code.

Tech: Vue.js, Nuxt.js, Node.js, Go, PostgreSQL, Docker, Podman, Kubernetes, AWS EKS, ArgoCD, Terraform, GitHub Actions

Software Engineering Intern | Sarawak Research Development Council, Kuching, Malaysia

Malaysian state government research agency

January 2024 to December 2024

[rpmn.sarawakrdc.org.my](#) (live internal system, login required)

- Collaborated with a 5-person team to build a grant application management system with React and TypeScript frontend, and a Node.js (Express) backend.
- Enabled real-time tracking of researcher profiles, publications, and performance metrics across departments.
- Deployed on DigitalOcean with Linux shell scripting and cron jobs for automated processing and backup.

Tech: React, TypeScript, Node.js, Express, EJS, SQL, DigitalOcean, Linux

EDUCATION

BSc Computer Science in Artificial Intelligence | Swinburne University of Technology, Sarawak Campus, Malaysia

Australian Bachelor's degree (AQF Level 7), awarded by Swinburne Melbourne

Relevant coursework: Machine Learning, Deep Learning, Data Structures and Algorithms, Software Engineering, Cloud Computing, Database Systems.

Activities: Vice President, Africa Unites Society (Swinburne).

2022 to 2025

Diploma in Business Information Technology | Asia Pacific University (APU), Kuala Lumpur, Malaysia

MQA-accredited private university in Malaysia

2019 to 2021

PERSONAL PROJECTS

Trainioapp | Personal Trainer and Client Marketplace App

Cross-platform mobile app, live on the App Store and Google Play

[App Store](#) | [Google Play](#)

Built a full-stack cross-platform mobile app connecting clients with personal trainers, featuring swipe-based trainer discovery, real-time chat, and session booking.

Implemented role-based flows for trainers and clients including profile management, certifications, earnings tracking, client leads, and workout/diet plan builder.

Integrated in-app subscriptions with RevenueCat and gated Pro features; published builds to Google Play and Apple App Store.

Tech: React Native, Expo, TypeScript, Supabase (PostgreSQL + Auth), RevenueCat, Google Maps API

August 2025 to March 2026

PupMood | Dog Mood Tracking Mobile App

Cross-platform mobile app, live on the App Store

[App Store](#)

Built a full-stack cross-platform mobile app for logging and analyzing a dog's mood over time, featuring multi-dog support, streak tracking, time-of-day mood bucketing, and animated UI with haptic and audio feedback.

Integrated Supabase for backend, authentication, and data sync, with Google and Apple Sign-In and a route-guarded onboarding flow.

Implemented in-app subscriptions with RevenueCat, gating multi-dog support, mood trends, and AI features behind entitlements.

Added AI-powered mood summaries that turn logged entries into natural-language insights, plus push notifications and reminders.

Handled the full release pipeline: EAS Build, App Store submission, and a marketing landing page with legal/support pages.

Tech: React Native, Expo, TypeScript, Supabase, RevenueCat, EAS Build

June 2026

MACHINE LEARNING PROJECTS (ACADEMIC)

Swinburne University of Technology, Sarawak Campus

2023 to 2024

Fall Detection CNN (2023): Trained a convolutional neural network in Python and TensorFlow to classify fall versus non-fall events from image data for elderly safety monitoring.

Face Anti-Spoofing Models (2023): Built computer vision models to separate live faces from photo and video presentation attacks, using facial texture, motion, and depth cues to harden a face recognition pipeline.

Football Sentiment Classifier (2024): Trained an LSTM text classifier on scraped Instagram comment data to label fan reactions as positive, negative, or neutral after match results.

Tech: Python, TensorFlow/Keras, OpenCV, NumPy, scikit-learn

TECHNICAL SKILLS

Languages and Frameworks: Python, TypeScript, JavaScript, Go, SQL, HTML/CSS, FastAPI, React.js, Next.js, Vue.js, Nuxt.js, Node.js, Express, React Native

Cloud and DevOps: AWS (EKS, S3, ALB), Kubernetes, Docker, Podman, Terraform, ArgoCD, GitHub Actions, CI/CD, Linux

Databases: PostgreSQL, SQL, Supabase, DigitalOcean managed DBs

AI and Machine Learning: RAG, vector search (pgvector), embeddings, prompt engineering, structured outputs, LLM integration, model training and evaluation, CNNs, LSTMs, computer vision, NLP, TensorFlow, OpenAI GPT-4o/GPT-5 API, KNIME Analytics

CERTIFICATIONS

- Machine Learning Specialization, DeepLearning.AI (June 2025)
- Kubernetes Hands-On: Deploy Microservices to AWS Cloud, Udemy (2025)

LANGUAGES

English: Native | German: A1 (in progress) | Dutch: Beginner

ACHIEVEMENTS & ACTIVITIES

- Vice President, Africa Unites Society, Swinburne University Sarawak (2023 to 2024).